Question	Answer	Marks
4(a)(i)	$v = r\omega$ $= (0.65)(540)$ $= 351\ \mathrm{m\ s}^{\ 1}$	 **[1]** **[1]**
4(a)(ii)	$a = r\omega^2$ $= (0.65)(540)^2$ $= 1.89 \cdot 10^5\ \mathrm{m\ s}^{\ 2} = 1.9 \cdot 10^5\ \mathrm{m\ s}^{\ 2}$	 **[1]** **[1]**
4(b)	Based on Newton's first law, the blade remains travelling in a straight line unless an external force is applied. Hence, there must be a force for the blade to move along the arc path of the turbine's wheel rim in a circular motion. From Newton's second law, this motion is facilitated by centripetal force which is provided by the tension in the attachment between the blade and the wheel of the turbine. At higher angular velocities for the wheel with the blades, larger centripetal force (tension) is needed. Hence, the blade breaks off when the tension reaches a maximum.	**[1]** **[1]** **[1]**
4(c)	Minimum radial force = ma $= (0.72)(1.89 \cdot 10^5)$ $= 1.36 \cdot 10^5\ \mathrm{N} = 1.4 \cdot 10^5\ \mathrm{N}$	 **[1]** **[1]**
	Max Marks	**9**

Question	Answer	Marks
5(c)	Since the standard cell is balanced, the p.d. across XY, V_{XY} = 1.02 V (i.e. e.m.f. of the standard	[1]